



Title: Implementation of PL/SQL stored procedure & functions.

Aim: To write and evaluate PL/SQL Stored procedure & function to perform a suitable task on the database & demonstrate their use.

Objective:

1. To understand the concept of PL/SQL procedure & functions.
2. To learn how to create & execute stored procedure.
3. To learn how to create use functions in SQL queries.
4. To differentiate between ~~procedure~~ procedures & function.
5. To apply PL/SQL in real database operations.

Theory :

1) Introduction :

PL/SQL (procedural language/structured query language) is Oracle's extension of SQL that allows procedural programming features like loops, conditions & functions.

It helps in writing efficient & reusable database programs.

PL/SQL supports:

- Stored procedures
- Functions
- Triggers.

2) Stored Procedures:

is a named block of PL/SQL code that help performs a specific task & can be stored in a database.

Syntax:

```
CREATE OR REPLACE PROCEDURE  
procedure_name (parameter)  
IS
```

BEGIN

END;

----- PL/SQL statements

Ex:

CREATE OR REPLACE PROCEDURE getSalary (empId Number)

IS

sal NUMBER

BEGIN

SELECT salary INTO sal FROM

employees WHERE id = emp-id;

DBMS-OUTPUT.PUT-LINE ('Salary' || sal);

END;

Execution:

BEGIN

get-salary (101);

END;

3) Function :

A function is similar to a procedure but always returns a value.

Syntax:
CREATE OR REPLACE Function function-name (parameters)
RETURN datatype
IS
BEGIN
RETURN value;
END;

Example:

```
CREATE OR REPLACE FUNCTION calculate-bonus  
(salary - NUMBER)  
RETURN NUMBER  
IS  
BEGIN  
RETURN salary * 0.10;  
END;
```

Usage:

```
SELECT calculate_bonus (50000) from dual;
```

* Characteristics of PL/SQL:

- Block structured language
- Supports Variable loops & conditions
- Improve performance by reducing network traffic.

- Error handling using EXCEPTION
- Code reusability through Procedure & Function

Diff. betⁿ. Procedure & Function

Procedure	Function
i) Procedure language does not return value	i) Function must return a value
ii) Called using EXEC or BEGIN --- END	ii) Usage can be used in SQL queries
iii) IN, OUT, IN OUT	iii) Only IN
iv) Perform actions	iv) Compute and return value
v) Ex: Data manipulation	v) Ex: Calculation

Conclusion: In this practical, we successfully created & executed a PL/SQL stored procedure & function. The procedure was used to perform a database operation while the function returned a calculated value. This demonstrated how PL/SQL enhances database functionality by enabling reusable efficiencies.

Timely Submission (5)	Journal Presentation (5)	Performance (5)	Understanding (5)	Oral (5)	Total (25)	Sign with Date
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